

Diet in acne: further evidence for the role of nutrient signalling in acne pathogenesis.



Recent evidence underlines the role of Western diet in the pathogenesis of acne. Acne is absent in populations consuming Palaeolithic diets with low glycaemic load and no consumption of milk or dairy products. Two randomized controlled studies, one of which is presented in this issue of *Acta Dermato-Venereologica*, have provided evidence for the beneficial therapeutic effects of low glycaemic load diets in acne. Epidemiological evidence confirms that milk consumption has an acne-promoting or acne-aggravating effect. Recent progress in understanding the nutrient-sensitive kinase mammalian target of rapamycin complex 1 (mTORC1) allows a new view of nutrient signalling in acne by both high glycaemic load and increased insulin-, IGF-1-, and leucine signalling due to milk protein consumption. Acne should be regarded as an mTORC1-driven disease of civilization, like obesity, type 2 diabetes and cancer induced by Western diet. Early dietary counselling of teenage acne patients is thus a great opportunity for dermatology, which will not only help to improve acne but may reduce the long-term adverse effects of Western diet on more serious mTORC1-driven diseases of civilization.